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Johann Bourcier (UPPA)

Subject: Application for PhD Position – Smart Scenario Exploration for Digital Twins through Validity Envelopes

Dear Johann Bourcier,

I am applying for the PhD position on "Smart Scenario Exploration for Digital Twins through Validity Envelopes" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

My experience in reinforcement learning and adversarial simulation has taught me the importance of defining validity envelopes when exploring AI-driven scenarios. Without strict bounding, generative exploration quickly leads to physically impossible states. I am highly motivated to apply my engineering background to design smart scenario algorithms that restrict digital twin simulations to mathematically and physically sound envelopes, particularly for environmental modeling.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui